Unbiased Estimators & Confidence Intervals

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Unbiased Estimator of Population Mean & Variance

A sample of size $n = 49$ has mean $\bar{x} = 44$, drawn from a population with $\mu_0 = 47$ and known $\sigma = 10$. Find the test statistic z, correct to 2 decimal places.

2 [Foundational] • Target: Confidence Intervals for Mean (Known Variance)

A sample of size $n = 49$ has mean $\bar{x} = 96$, drawn from a population with $\mu_0 = 98$ and known $\sigma = 12$. Find the test statistic z, correct to 2 decimal places.

3 [Foundational] • Target: Confidence Intervals for Mean (Unknown Variance, t-distribution)

A sample of size $n = 25$ has mean $\bar{x} = 76$, drawn from a population with $\mu_0 = 80$ and known $\sigma = 8$. Find the test statistic z, correct to 2 decimal places.

4 [Foundational] • Target: Interpretation of Confidence Levels

A sample of size $n = 25$ has mean $\bar{x} = 62$, drawn from a population with $\mu_0 = 59$ and known $\sigma = 10$. Find the test statistic z, correct to 2 decimal places.

5 [Foundational] • Target: Unbiased Estimator of Population Mean & Variance

A sample of size $n = 25$ has mean $\bar{x} = 51$, drawn from a population with $\mu_0 = 54$ and known $\sigma = 5$. Find the test statistic z, correct to 2 decimal places.

6 [Foundational] • Target: Confidence Intervals for Mean (Known Variance)

A sample of size $n = 49$ has mean $\bar{x} = 87$, drawn from a population with $\mu_0 = 85$ and known $\sigma = 5$. Find the test statistic z, correct to 2 decimal places.

7 [Foundational] • Target: Confidence Intervals for Mean (Unknown Variance, t-distribution)

If $X \sim N(\mu, \sigma^2)$, approximately what percentage of values lie within 3 standard deviations of the mean? Show your complete algebraic work.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Interpretation of Confidence Levels

If $X \sim N(\mu, \sigma^2)$, approximately what percentage of values lie within 2 standard deviations of the mean? Show your complete algebraic work.

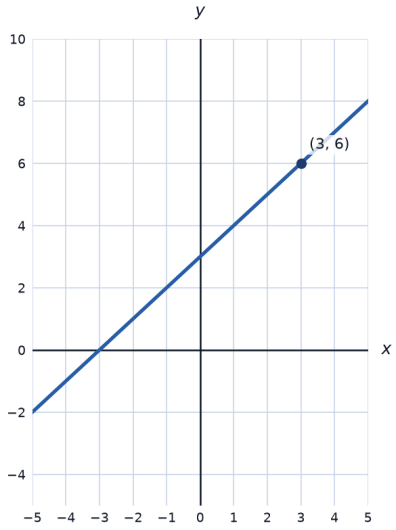
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9 **[Intermediate]** • **Target: Unbiased Estimator of Population Mean & Variance**

If $X \sim N(\mu, \sigma^2)$, approximately what percentage of values lie within 1 standard deviation of the mean? Show your complete algebraic work.

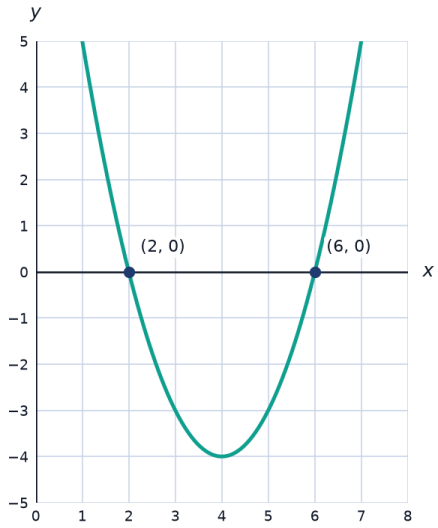
10 **[Intermediate]** • **Target: Confidence Intervals for Mean (Known Variance)**

Consider the function $f(x) = \frac{x^2 - 9}{x - k}$. The graph of $y = f(x)$ has a removable discontinuity at $x = 3$. Find the value of k. The graph of $y = f(x)$ is shown below.



11 **[Intermediate]** • **Target: Confidence Intervals for Mean (Unknown Variance, t-distribution)**

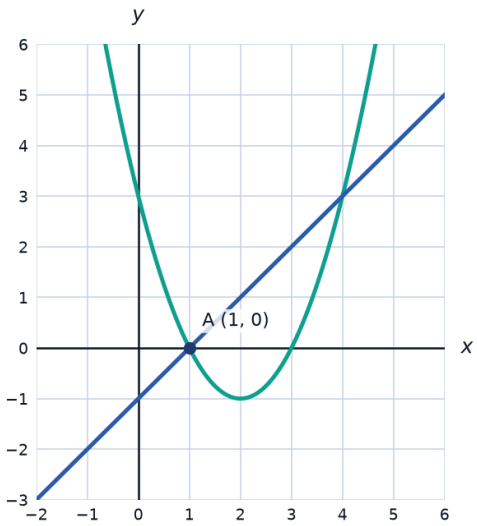
Consider the quadratic function $f(x) = x^2 - 8x + k$. The graph of $y = f(x)$ is a parabola that intersects the x-axis at the points $(2, 0)$ and $(6, 0)$. Find the value of k.



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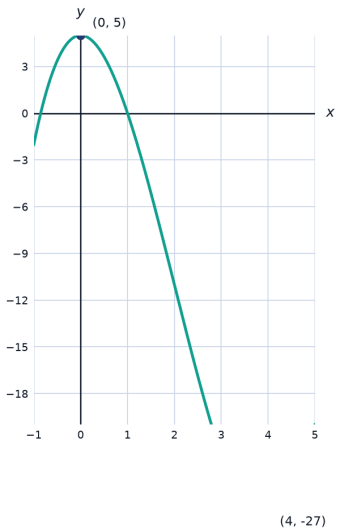
12 **[Intermediate]** • **Target: Interpretation of Confidence Levels**

The diagram shows the graph of the function $f(x) = x^2 - 4x + 3$ and a line $y = g(x)$. The line $y = g(x)$ intersects the parabola $y = f(x)$ at points A and B. Point A is given as $(1, 0)$. Find the x-coordinate of point B.



13 **[Intermediate]** • **Target: Unbiased Estimator of Population Mean & Variance**

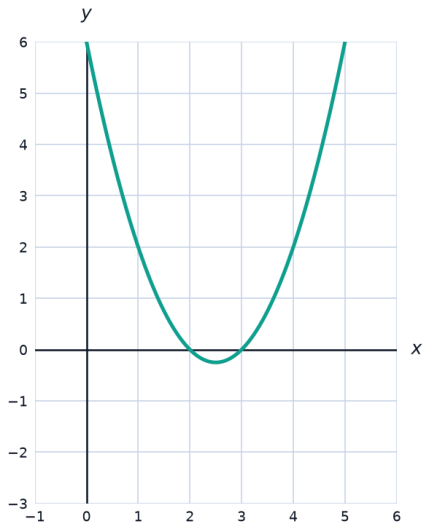
The graph of the function $f(x) = x^3 - 6x^2 + 5$ is shown below. The graph has a local maximum and a local minimum. Find the x-coordinate of the local minimum.



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14 [Intermediate] • Target: Confidence Intervals for Mean (Known Variance)

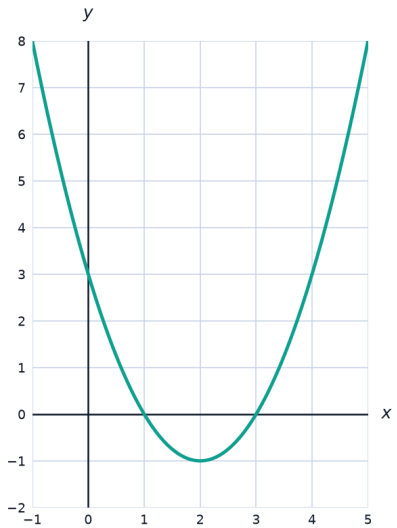
Consider the function $f(x) = x^2 - 5x + 6$. The graph of this function is a parabola. Determine the x-coordinates of the points where the graph intersects the x-axis.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Confidence Intervals for Mean (Unknown Variance, t-distribution) • Cross-Lesson: 3.01 3D Vectors & the Cross Product

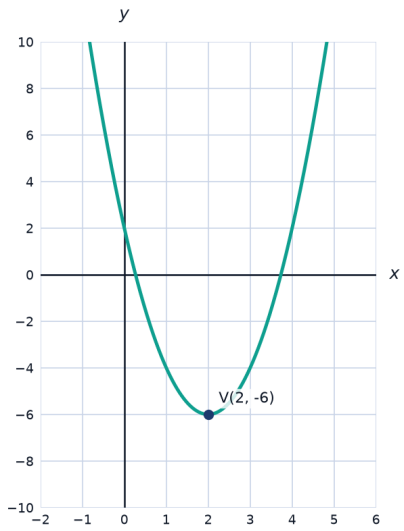
The diagram shows the graph of the function $y = f(x)$. The equation of the function is given by $f(x) = x^2 - 4x + 3$. Identify the coordinates of the local minimum of the graph.



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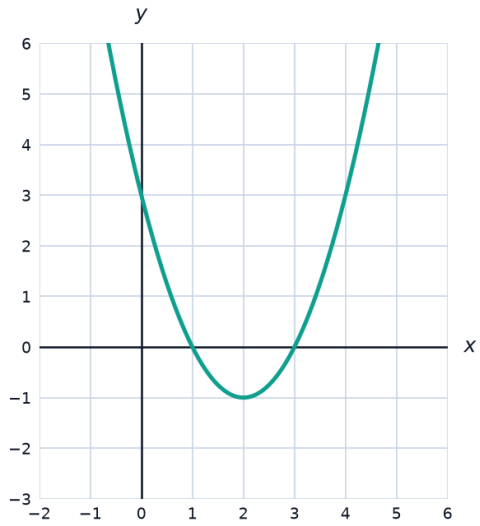
16 **[Challenging]** • **Target: Interpretation of Confidence Levels** • **Cross-Lesson: 3.02 Lines and Planes in Three Dimensions**

The diagram shows the graph of the function $f(x) = 2x^2 - 8x + k$. The graph intersects the x-axis at points A and B, and the vertex of the parabola is at point V. Given that the x-coordinate of the vertex is 2, and the y-coordinate of the vertex is -6, find the value of k.



17 **[Challenging]** • **Target: Unbiased Estimator of Population Mean & Variance** • **Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)**

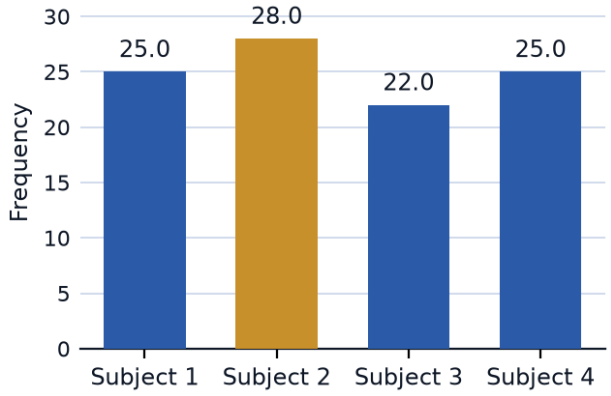
The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at points A and B. Determine the x-coordinate of the vertex of the parabola.



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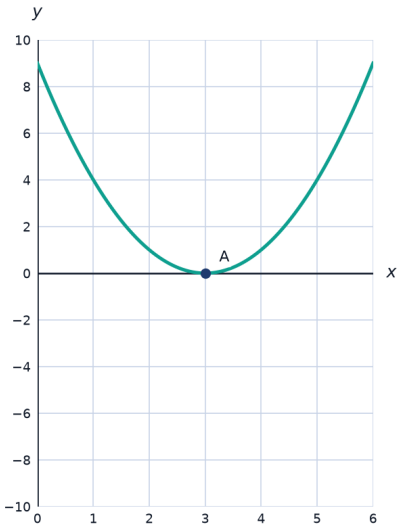
18 [Challenging] • Target: Confidence Intervals for Mean (Known Variance) • Cross-Lesson: 4.02 The Poisson Distribution

The following bar chart shows the number of students who chose each of four different subjects in a school. Let N be the total number of students surveyed. Find the value of N.



19 [Challenging] • Target: Confidence Intervals for Mean (Unknown Variance, t-distribution) • Cross-Lesson: 5.01 Further Integration Techniques

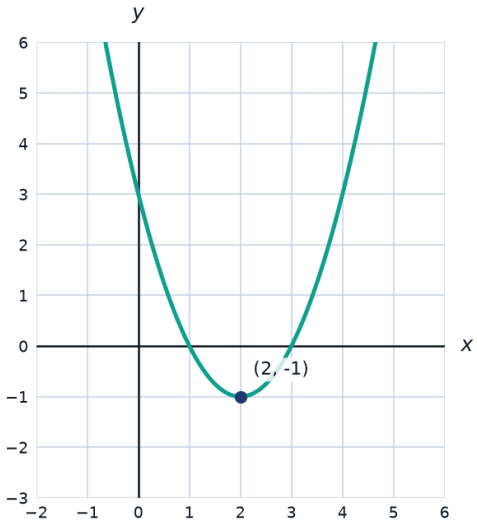
The diagram shows the graph of a quadratic function $y = f(x) = x^2 - 6x + k$. The graph intersects the x-axis at point A, and has a minimum point at B. The coordinates of point A are $(3, 0)$. Find the value of k.



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20 [Challenging] • Target: Interpretation of Confidence Levels • Cross-Lesson: 5.02 Differential Equations

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is a parabola. Find the coordinates of the vertex of the parabola.



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Unbiased Estimator of Population Mean & Variance

Given sample observations $x_1, \ldots, x_n$, what is the formula for the unbiased estimator s^2_{n-1} of the population variance σ^2?

a. $s^2_{n-1} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$

b. $s^2_n = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$

c. $s^2 = \frac{1}{n+1}\sum_{i=1}^{n}x_i^2$

d. $s^2 = \frac{(\sum x_i)^2}{n-1}$

22 [Foundational] • Target: Unbiased Estimator of Population Mean & Variance

A sample of $n = 5$ values has $\sum x_i = 30$ and $\sum x_i^2 = 210$. Calculate the unbiased estimate of the population variance s^2_{n-1}.

a. 7.5

b. 6.0

c. 30

d. 9.0

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23 [Foundational] • Target: Unbiased Estimator of Population Mean & Variance

If a sample standard deviation computed with denominator n is $s_n = 4.0$ for $n = 16$, what is the unbiased standard deviation s_{n-1}?

a. $s_{n-1} = 4.0 \times \sqrt{\frac{16}{15}} \approx 4.13$

b. $s_{n-1} = 4.0 \times \sqrt{\frac{15}{16}} \approx 3.87$

c. $s_{n-1} = 4.0$

d. $s_{n-1} = 4.0 \times \frac{16}{15} \approx 4.27$

24 [Foundational] • Target: Confidence Intervals for Mean (Known Variance)

A random sample of $n = 64$ items from a population with known $\sigma = 12$ has mean $\bar{x} = 50$. What is the 95% confidence interval for μ ($z^* = 1.96$)?

a. $[47.06, 52.94]$

b. $[48.50, 51.50]$

c. $[44.12, 55.88]$

d. $[49.63, 50.37]$

25 [Foundational] • Target: Confidence Intervals for Mean (Known Variance)

To reduce the margin of error of a confidence interval by a factor of 3, the sample size must be multiplied by:

a. 9

b. 3

c. $\sqrt{3}$

d. 6

26 [Foundational] • Target: Confidence Intervals for Mean (Known Variance)

What is the critical value z^* for a 99% confidence interval?

a. 2.576

b. 1.960

c. 1.645

d. 2.326

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27 [Foundational] • Target: Confidence Intervals for Mean (Unknown Variance, t-distribution)

When constructing a confidence interval for μ with unknown population variance σ^2 from a normal sample of size $n = 10$, which distribution is used?

a. Student's t-distribution with 9 degrees of freedom

b. Standard Normal Z-distribution

c. Student's t-distribution with 10 degrees of freedom

d. Chi-square distribution with 9 degrees of freedom

28 [Foundational] • Target: Confidence Intervals for Mean (Unknown Variance, t-distribution)

How does the critical value t^* compare to z^* for the same confidence level and finite sample size?

a. $t^* > z^*$, because the t-distribution has heavier tails to reflect estimation uncertainty

b. $t^* < z^*$, because t is more compact

c. $t^* = z^*$ always

d. t^* can be smaller or larger depending on $\bar{x}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Confidence Intervals for Mean (Unknown Variance, t-distribution)

A sample of $n = 16$ has $\bar{x} = 22.0$ and $s_{n-1} = 2.0$. Given $t_{0.025,\,15} = 2.131$, find the 95% confidence interval for μ.

a. $[20.93, 23.07]$

b. $[21.47, 22.53]$

c. $[19.87, 24.13]$

d. $[20.00, 24.00]$

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30 **[Intermediate]** • **Target: Interpretation of Confidence Levels**

Which statement correctly conveys the meaning of a '95% confidence level'?

a. If many random samples are drawn and intervals constructed, 95% of them will contain the true population parameter

b. There is a 95% probability that the true parameter lies inside the specific interval $[47, 53]$

c. 95% of sample data values lie between 47 and 53

d. The sample mean has a 95% chance of being correct

31 **[Intermediate]** • **Target: Interpretation of Confidence Levels**

If a 95% confidence interval for μ is $[102, 118]$, what is the conclusion for the two-tailed test $H_0 : \mu = 100$ vs $H_1 : \mu \neq 100$ at $\alpha = 0.05$?

a. Reject H_0 at the 5% level, because 100 lies outside the 95% confidence interval

b. Fail to reject H_0, because 100 is close to 102

c. Accept H_0 as proven

d. The test is inconclusive without a P-value

32 **[Intermediate]** • **Target: Interpretation of Confidence Levels**

What happens to the width of a confidence interval if the confidence level is increased from 90% to 99% (with sample size fixed)?

a. The interval widens

b. The interval narrows

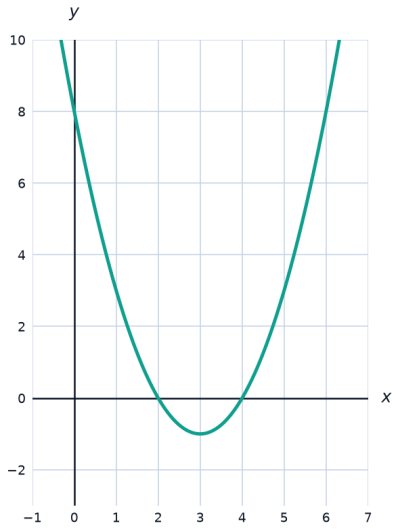
c. The interval width remains identical

d. The interval shifts to the right

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33 [Intermediate] • Target: Unbiased Estimator of Population Mean & Variance

Consider the function $f(x) = x^2 - 6x + 8$. The graph of $y = f(x)$ intersects the x-axis at points A and B, and the y-axis at point C. Find the length of the line segment AB.



a. 2 units

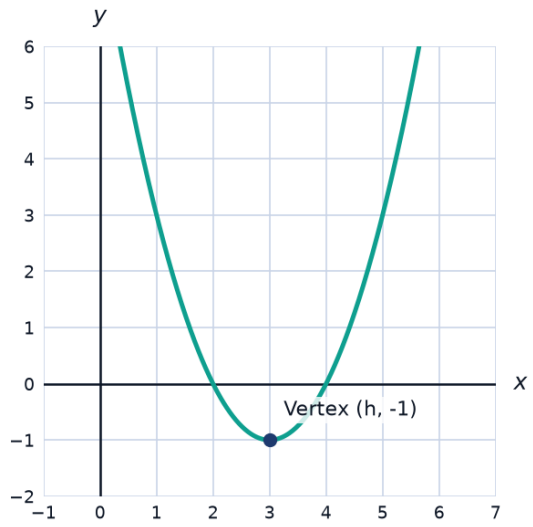
b. 3 units

c. 4 units

d. 6 units

34 [Intermediate] • Target: Confidence Intervals for Mean (Known Variance)

The diagram shows the graph of the function $f(x) = x^2 - 6x + k$. The vertex of the parabola is at the point $(h, -1)$. Determine the value of k.



a. 8

b. 7

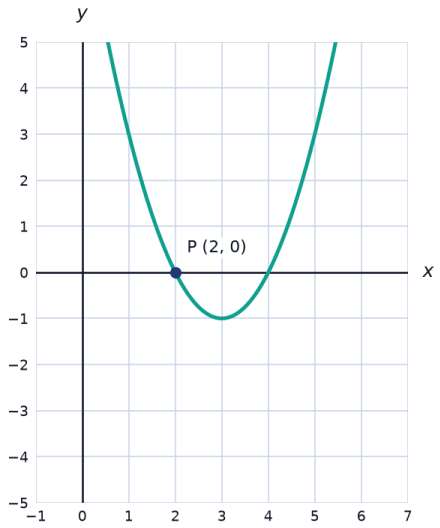
c. 6

d. 5

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35 **[Intermediate]** • **Target: Confidence Intervals for Mean (Unknown Variance, t-distribution)**

The graph of the function $g(x) = x^2 - 6x + k$ is shown below. The graph intersects the x-axis at points P and Q. Point P has coordinates $(2, 0)$. Find the value of the parameter k.



a. 1

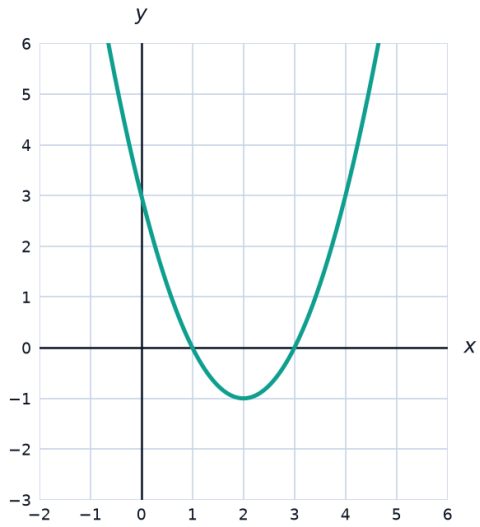
b. 2

c. 4

d. 8

36 **[Intermediate]** • **Target: Interpretation of Confidence Levels**

Consider the graph of the quadratic function $y = x^2 - 4x + 3$. What are the coordinates of the vertex of this parabola?



a. (2, -1)

b. (1, 0)

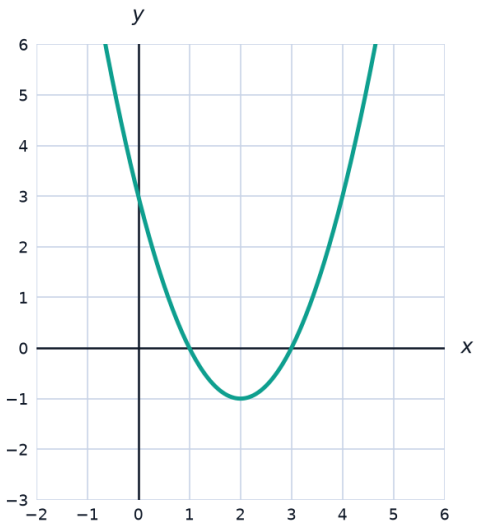
c. (3, 0)

d. (2, 1)

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37 [Intermediate] • Target: Unbiased Estimator of Population Mean & Variance

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown below. The graph intersects the x-axis at points A and B. Determine the distance between A and B.



a. 2

b. 3

c. 4

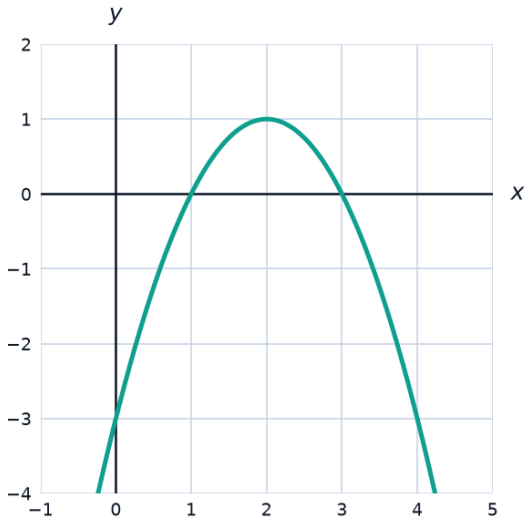
d. 5

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Confidence Intervals for Mean (Known Variance) • Cross-Lesson: 4.02 The Poisson Distribution

The graph of the function $f(x) = -x^2 + 4x - 3$ is shown below. Determine the value of $f'(2)$.



a. -2

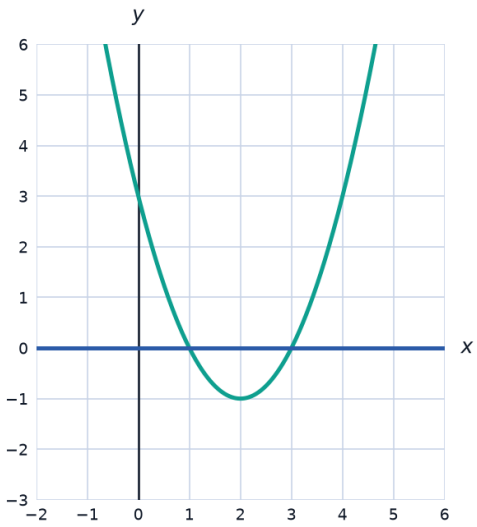
b. 0

c. 2

d. 4

39 [Challenging] • Target: Confidence Intervals for Mean (Unknown Variance, t-distribution) • Cross-Lesson: 5.01 Further Integration Techniques

Consider the quadratic function $f(x) = x^2 - 4x + 3$. Find the value of x where the graph of $f(x)$ intersects the x-axis at the point $(x, 0)$, given that $x > 1$.



a. 2

b. 3

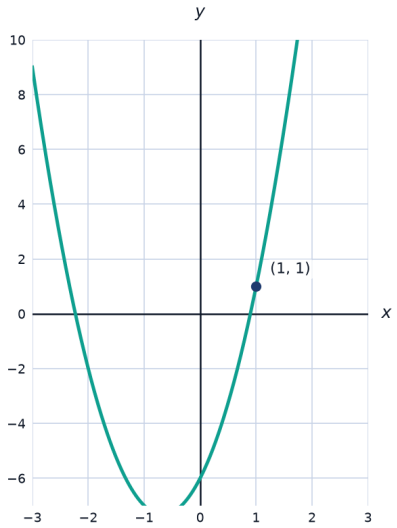
c. 4

d. 5

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40 [Challenging] • Target: Interpretation of Confidence Levels • Cross-Lesson: 5.02 Differential Equations

The diagram shows the graph of the function $f(x) = 3x^2 + bx - 6$. The graph passes through the point $(1, 1)$. Find the value of b.



a. 3

b. 4

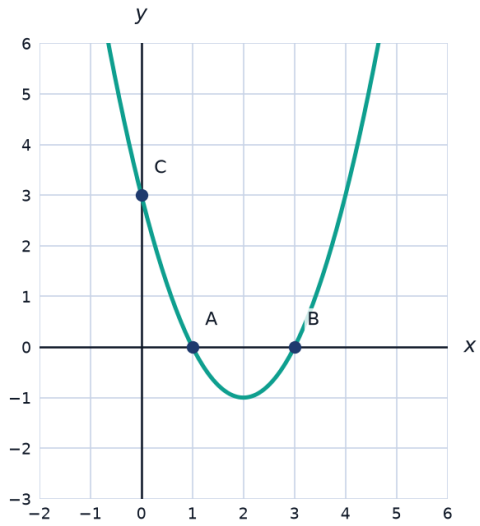
c. 5

d. 6

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41 **[Challenging]** • **Target: Unbiased Estimator of Population Mean & Variance** • **Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series**

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown below. The graph intersects the x-axis at points A and B, and the y-axis at point C. Find the coordinates of the point where the graph of $y = f(x)$ intersects the y-axis.



a. (0, 3)

b. (3, 0)

c. (1, 0)

d. (0, -3)

42 **[Challenging]** • **Target: Confidence Intervals for Mean (Known Variance)** • **Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)**

A particle's position in the complex plane is given by $z(t) = (2t^2 - 3t + 1) + i(t^3 - 6t^2 + 11t - 6)$, where t is time in seconds and $t \geq 0$. The particle starts at $t = 0$ and its path is traced by the complex number $z(t)$. The particle is said to be 'at rest' when its velocity, represented by the complex number $v(t) = z'(t)$, is zero. Determine the times $t > 0$ when the particle is at rest.

a. $t = 1$ and $t = 3$

b. $t = 1$ and $t = 2$

c. $t = 2$ and $t = 3$

d. $t = 1, t = 2,$ and $t = 3$

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43 [Challenging] • Target: Confidence Intervals for Mean (Unknown Variance, t-distribution) • Cross-Lesson: 6.01 Matrices (AI HL)

Consider the function $f(x) = \frac{ax^2 + bx + c}{x - d}$, where a, b, c, and d are constants. The graph of $f(x)$ has a vertical asymptote at $x = 3$ and an oblique asymptote $y = 2x + 1$. Given that $f(1) = -4$ and $f(2) = -6$, find the value of a.

a. -2

b. 2

c. -4

d. 4

44 [Challenging] • Target: Interpretation of Confidence Levels • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

Consider the complex number w such that $w = x + yi$, where x and y are real numbers. Given that $w^3 = -64i$ and the imaginary part of w is positive, find the value of $x^2 + y^2$.

a. 16

b. 32

c. 64

d. 8

45 [Challenging] • Target: Unbiased Estimator of Population Mean & Variance • Cross-Lesson: 6.03 Markov Chains (AI HL)

Consider the function $f(x) = ax^2 + bx + c$. The graph of $f(x)$ is a parabola that passes through the points $(-1, 12)$, $(1, 2)$, and $(2, 9)$. Find the value of the parameter b.

a. -3

b. 3

c. -5

d. 5

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